

BANARAS HINDU UNIVERSITY
Midterm Examination 2025-2026

Program: B.Sc.

Year: II

Semester: II

Subject: Mathematics

Subject Code: MATMJ44 (Mechanics)

Date: 26/03/2026

Start Time: 9:00 AM

Time Duration: One Hour

Max. Marks: 20

Instruction: 1) Answer any Four questions. The figures on the right hand margin indicate marks.

2) Write your section on top right of the front page of your answer script.

3) Give proper justification for your answers. Marks will not be awarded for guess work.

Question 1.

- (a) Define concurrent forces. Find analytical expression for resultant of two concurrent forces. [3]
- (b) The angle of inclination between two forces P and Q is θ . If P and Q be interchanged in position, show that the resultant will be turned through an angle ϕ , where $\tan \frac{\phi}{2} = \frac{P-Q}{P+Q} \tan \frac{\theta}{2}$, ($P > Q$). [2]

Question 2.

- (a) State and prove the necessary and sufficient conditions for the equilibrium of a system of coplanar forces acting on a rigid body. [5]

Question 3.

- (a) Define like parallel forces and obtain resultant of two like parallel forces. The resultant of two like parallel forces P and Q passes through a point O ; when P is increased by R and Q by S , the resultant still passes through O , and also when Q, R replace P, Q , respectively; show that $S = R - \frac{(Q-R)^2}{P-Q}$. [5]

Question 4.

- (a) Define moment of a force about a point. Show that the algebraic sum of moments of two parallel unlike forces about any point in their plane is equal to the moment of their resultant about that point. [5]

Question 5.

- (a) Show that a system of particles at rest when acted upon by a system of coplanar forces remains in equilibrium if and only if the resultant work done by the forces is zero. [5]